whitenoise

A Python Package for Stochastic White Noise Analysis (SWNA) with Memory

1. What the Package Does

whitenoise is a research software package that quantifies *memory* in fluctuating time-series data — that is, whether past fluctuations in a system influence its future behavior, and how strongly. It implements Stochastic White Noise Analysis (SWNA), a framework in which an observable is modeled as a stochastic integral driven by white noise and shaped by a memory kernel:

$$x(T) = x_0 + \int_0^T f(T-t) \cdot h(t) \cdot \omega(t)\, dt$$

Different choices of the memory kernel `f·h` correspond to different physical memory behaviors (oscillatory, exponential, self-similar, restricted, etc.). The package fits these theoretical kernels to the **Mean Square Displacement (MSD)** computed from real data, extracting a **memory parameter μ** together with 95% confidence intervals. The magnitude and structure of μ indicate how strongly a system "remembers" its own past — information not directly visible from the raw time series.

The package accepts *any* sequential one-dimensional observable — not only time series in the literal sense. The independent variable can be time, an occurrence index, a genomic base-pair position, or any other ordered quantity, provided it is stored as a two-column CSV file (`independent_variable [unit], observable [unit]`).

2. Analysis Pipeline

The package runs one consistent pipeline regardless of the dataset or model chosen:

1. **Read** — parse the CSV, auto-detect column names/units, run automatic data-quality checks (e.g. possible column swap, unexpected negative values).
2. **Preprocess** *(optional)* — detrend (linear/polynomial/moving-average/power-spectral-density/cycle-averaging), normalize, or smooth the series to isolate fluctuations from deterministic trends.
3. **Compute MSD** — the empirical Mean Square Displacement is calculated at each time lag.
4. **Fit** — a theoretical MSD formula is least-squares fitted to the empirical MSD, using a dual-fit strategy (with and without a free normalization scalar) so the better-fitting formulation is automatically reported.
5. **Report** — fitted parameters, standard errors, 95% confidence intervals, and goodness-of-fit (R^2) are returned, printed, and optionally exported to CSV.
6. **Visualize** — diagnostic and publication-quality plots (raw series, MSD fit overlay, displacement probability distribution) are generated automatically.

Model selection is always explicit and researcher-controlled — the package never silently auto-selects a model. A batch mode can try every available model on one dataset and rank the results by goodness-of-fit, to help narrow down which memory behavior best matches new data.

3. Theoretical Models Implemented

The package's model registry currently contains **19 theoretical MSD formulas** drawn from the SWNA reference literature. **10 are fully implemented** and ready to fit; the remaining 9 are registered as placeholders for future implementation (the package reports their status transparently rather than silently omitting them).

Model	Parameters	Behavior captured

Exponential	μ, β	Monotonically growing MSD with exponential damping — fast-rise, memory-driven processes.
Cosine / Sine	μ, ν	Oscillatory MSD growth — periodic or quasi-periodic systems.
Fractional Brownian Motion	H (Hurst exponent)	Self-similar processes; H = 0.5 recovers ordinary Brownian motion.
Restricted diffusion (plateau)	a, b, c	MSD that saturates at a plateau rather than growing indefinitely — bounded/confined fluctuations.
Incomplete-gamma, Bessel-function forms, ...	model-specific	Additional memory shapes for specialized fluctuation behaviors.

For the oscillatory models, the short-time behavior reduces to a power law (MSD ≈ $c{\cdot}T^{\alpha}$, with α = 2μ−1), which the package uses to classify the fitted result along a standard diffusive spectrum (sub-diffusive / Brownian / super-diffusive / hyperballistic). For the exponential model, whose MSD does not reduce to a clean power law, the package instead classifies μ directly as a memory-persistence indicator (memoryless / long-memory / short-memory). The two schemes are kept strictly separate and are only applied to the model family each is derived for — the package does not apply a single generic threshold table across all models.

4. Validated Applications

The fitting pipeline has been validated by reproducing memory-parameter estimates for several real, published datasets spanning different scientific domains:

Domain	Dataset type	Model used	Result range
Seismology	Earthquake inter-event times	Exponential	μ ≈ 1.00–1.19
Solar physics	Sunspot number cycles	Exponential	μ ≈ 1.15
Marine ecology	Coral reef bleaching records	Cosine	μ ≈ 4.64
Atmospheric science	Atmospheric CO_2 concentration (Keeling curve)	Cosine	μ ≈ 0.91–0.97
Genomics	Nucleotide-separation distances in bacterial genome sequences	Restricted diffusion (plateau)	plateau ≈ 5.2
Radio astronomy	Solar/Jovian radio burst intensity fluctuations	Exponential	μ ≈ 1.4–1.6

5. Package Structure and Extensions

The core package (`whitenoise`) provides the general-purpose pipeline described above, along with an automated test suite (78 tests covering data I/O, MSD computation, model fitting, analysis orchestration, comparison/batch tools, and visualization). Two domain-specific extensions are built on top of the same core engine:

Genomics extension

Applies SWNA to DNA sequence data by treating the distance between consecutive occurrences of a nucleotide pair (e.g. A→A, A→T) along a genome as the fluctuating observable. Includes FASTA file reading, nucleotide transition-distance extraction for all 16 base-pair combinations, and a dedicated restricted-diffusion model fit to characterize genomic memory structure.

Radio astronomy extension (RadioJOVE)

Applies SWNA to radio burst observations (e.g. solar Type III bursts, Jovian S-bursts) recorded with amateur/citizen-science radio spectrographs. Provides burst-specific CSV parsing, cadence-aware batch analysis of multiple burst regions, and comparison tools to contrast memory behavior across bursts recorded at different times or stations.

6. Key Capabilities Summary

Capability	Description
Input	Any 2-column (or multi-column, for batch mode) CSV with a self-describing `name [unit]` header; automatic unit-aware axis labeling and data-quality warnings.
Preprocessing	5 detrending methods, 4 normalization methods, 2 smoothing methods — all optional and manually invoked.
Fitting	Nonlinear least-squares fitting with automatic dual-mode comparison, 95% confidence intervals, and goodness-of-fit reporting for every parameter.
Multi-dataset comparison	Side-by-side comparison of memory parameters across multiple datasets or multiple candidate models, with ranked summary tables.
Batch processing	Whole-folder or multi-column-file batch analysis, with optional parallel execution.
Visualization	Exploratory (fast, annotated) and publication-quality (styled, exportable) plotting for raw series, MSD fits, and displacement distributions.
Export	CSV export of fitted MSD curves, comparison summaries, and checkpointed intermediate data at any pipeline stage.
Extensibility	A shared core pipeline reused unmodified by both domain-specific extensions (genomics, radio astronomy), demonstrating the framework's applicability beyond its original design domain.

whitenoise v0.1.0 · Python ≥ 3.10 · open-source (NumPy, SciPy, pandas, Matplotlib)